

Multilingual AI Customer Support Bot — Workflow Documentation

Overview

This workflow automates customer support using AI agents inside n8n. It processes incoming emails, understands user intent, and either:

-  Replies automatically
-  Escalates to a human

It supports **multiple languages** and responds in the **user's original language**.

Core Capabilities

-  Language detection (Hindi, Telugu, English, etc.)
 -  Translation to English for processing
 -  Intent classification (Refund, Technical Issue, Complaint, Pricing)
 -  AI-generated responses
 -  Decision engine (Auto Reply vs Escalation)
 -  Multilingual reply (same language as input)
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Workflow Architecture

```
Gmail Trigger
  ↓
Normalize Input
  ↓
Language Detection Agent
  ↓
Confidence Check
  ├── Low Confidence → Fallback (English)
  └── High Confidence → Continue
  ↓
Translation to English
  ↓
Classification Agent
  ↓
Decision Engine (IF Node)
  ├── Escalation Flow
  └── Auto Reply Flow
```

Detailed Node Breakdown

1. Gmail Trigger

- Listens for incoming emails
 - Extracts:
 - message
 - subject
 - sender email
-

2. Normalize Input (Set Node)

Standardizes input fields:

```
{  
  "message": "...",  
  "email": "...",  
  "subject": "...",  
  "start_time": "timestamp"  
}
```

3. Language Detection Agent

Detects:

- `language` (e.g., hi, te, en)
 - `confidence` (0-1)
-

4. Confidence Check (IF Node)

Condition:

- `confidence < threshold` (e.g., 0.6)

Outcome:

- TRUE → fallback to English
 - FALSE → continue flow
-

5. Translation to English

- Converts user input → English
 - Ensures consistent processing for AI
-

6. Classification Agent

Returns:

```
{
  "category": "Refund | Technical Issue | Complaint | Pricing",
  "confidence": 0.0-1.0,
  "urgency": "Low | Medium | High"
}
```

7. Decision Engine

Condition:

- urgency = High
OR
- classification confidence < threshold

Escalation Flow

Restore Context

Reattach:

- message
- email
- subject

Add Escalation Metadata

```
{
  "status": "Escalated",
  "reason": "High urgency or low confidence"
}
```

Send Escalation Email

- Sent to admin/support team
 - Includes full user context
-

Auto Reply Flow

Restore Context

Ensure all fields are available:

- message
 - email
 - subject
 - language
-

Reply Agent

Generates:

- short, professional response
 - based on category
-

Translation to User Language

- Converts reply → original language
 - Ensures seamless user experience
-

Send Reply Email

- Sent to user
 - Uses translated response
-

Data Flow Summary

User Email → Detect Language → Translate → Classify → Decide → Reply/Escalate

Example Flow

Input (Hindi)

मुझे मेरा पैसा वापस चाहिए

Processing

- Language → Hindi

- Translation → "I want my money back"
- Category → Refund

Output (Hindi)

आपकी रिफंड अनुरोध स्वीकार कर लिया गया है।

⚠ Edge Cases

Scenario	Handling
Low confidence language	Fallback to English
Multiple intents	May misclassify
Mixed languages	Reduced accuracy

🔧 Configuration Requirements

Required Credentials:

- Gmail API
- OpenAI / LLM API

📊 Performance Considerations

- Response time depends on API latency
- Translation accuracy varies by language
- Best performance with clear single queries

🌐 Future Enhancements

- 📊 Logging (Google Sheets / DB)
- ⌚ Response time tracking
- 📈 Analytics dashboard
- 🌐 Expanded language support
- 📱 WhatsApp / chat integrations

Conclusion

This workflow demonstrates a **scalable AI-driven support system** capable of:

- Reducing manual workload
 - Improving response time
 - Supporting global users
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